

A Guide to the Excel Data Analysis Toolpack: Linear Regression Output

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Let's assume our regression model is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \epsilon$$

Where:

- y is the dependent variable.
- $x_1, \dots, x_k$ are the independent (predictor) variables.
- k is the number of independent variables.
- β_0 is the intercept.
- $\beta_1, \dots, \beta_k$ are the coefficients for the predictors.
- ϵ is the error term.

And n will be our sample size (number of observations).

1 Section 1: Regression Statistics

This table provides a summary of the model's overall fit.

- **Multiple R:** The correlation coefficient, R . It measures the strength of the linear relationship between the predicted values ($\hat{y}$) and the actual values (y). For SLR It is the positive square root of R Square.
- **R Square (R^2):** The Coefficient of Determination. This is a key metric of model fit. It represents the proportion of the total variance in the dependent variable (y) that is explained by the independent variable(s) (x). It is calculated as

$$R^2 = \frac{\text{Explained Variance}}{\text{Total Variance}} = \frac{SS_{reg}}{SS_{total}} = 1 - \frac{SS_{residual}}{SS_{total}}$$

- **Adjusted R Square:** A modified version of R^2 that adjusts for the number of predictors (k) in the model. It is generally preferred for multiple regression (when $k > 1$) because it penalizes the model for adding extra variables that don't significantly improve the fit.
- **Standard Error:** This is the **Standard Error of the Regression** (also called the Root Mean Square Error or RMSE). It represents the standard deviation of the residuals (ϵ). It gives you a sense of the average distance that the observed values fall from the regression line. It is directly related to MS_{error} from the ANOVA table.
- **Observations:** The total number of data points in your sample, n .

2 Section 2: ANOVA (Analysis of Variance)

The ANOVA table tests the overall significance of the regression model. It partitions the total variation in y into two parts: the variation explained by the regression (Regression) and the unexplained variation (Residual).

Hypothesis Test for Overall Model Significance

- **Null Hypothesis (H_0):** The model is not significant. All slope coefficients are zero.

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0$$

- **Alternative Hypothesis (H_a):** The model is significant. At least one slope coefficient is not zero.

$$H_a : \text{At least one } \beta_j \neq 0 \text{ (for } j = 1, \dots, k)$$

ANOVA Table Breakdown

Term	Formula	Description
df (Degrees of Freedom)		
Regression	$df_{reg} = k$	Number of independent variables.
Residual	$df_{error} = n - k - 1$	Sample size minus number of all parameters.
Total	$df_{total} = n - 1$	Sample size minus 1.
SS (Sum of Squares)		
Regression	$SS_{reg} = \sum(\hat{y}_i - \bar{y})^2$	Variation explained by the model.
Residual	$SS_{error} = \sum(y_i - \hat{y}_i)^2$	Unexplained variation (residuals).
Total	$SS_{total} = \sum(y_i - \bar{y})^2$	Total variation in y .
MS (Mean Square)		
Regression	$MS_{reg} = \frac{SS_{reg}}{df_{reg}}$	Average explained variation.
Residual	$MS_{error} = \frac{SS_{error}}{df_{error}}$	Average unexplained variation.
F-Statistic		
F	$F = \frac{MS_{reg}}{MS_{error}}$	The test statistic for overall significance.
Significance F		
p-value	The p-value associated with the F-statistic. If this value is less than your chosen alpha level (e.g., $\alpha = 0.05$), you reject H_0 and conclude that your model is statistically significant.	

Relation Between MS(error) and Standard Error

The relationship between the **Standard Error** (from the Regression Statistics section) and the **MS(error)** (from the ANOVA table) is as follows.

The **MS(error)** is the **variance** of the residuals (the average squared error). The **Standard Error** of the regression is the **standard deviation** of the residuals.

Therefore, the relationship is:

$$\text{Standard Error} = \sqrt{MS_{error}}$$

3 Section 3: Coefficients Table

This table provides information about each individual predictor (and the intercept) in the model.

Hypothesis Test for Individual Coefficients

For each coefficient β_j (e.g., Intercept, β_1 , β_2 , ...), a separate t-test is performed:

- **Null Hypothesis (H_0):** The coefficient is not statistically significant (it is equal to zero).

$$H_0 : \beta_j = 0$$

- **Alternative Hypothesis (H_a):** The coefficient is statistically significant (it is not equal to zero).

$$H_a : \beta_j \neq 0$$

Coefficients Table Breakdown

- **Coefficients:** These are the estimated values for β_0 (Intercept) and $\beta_1, \dots, \beta_k$ (slopes).
- **Standard Error:** The standard error for each individual coefficient, $SE(\beta_j)$. It measures the uncertainty in the estimation of that coefficient.
- **t Stat:** The test statistic for the individual coefficient. It is calculated by:

$$t = \frac{\text{Coefficient} - 0}{\text{Standard Error}} = \frac{\beta_j}{SE(\beta_j)}$$

- **P-value:** The p-value associated with the t-statistic. If this value is less than your alpha level (e.g., $\alpha = 0.05$), you reject H_0 for that specific coefficient and conclude it is a statistically significant predictor.
- **Lower 95% / Upper 95%:** This is the 95% confidence interval for the true value of the coefficient. If this interval **does not** contain 0, it is equivalent to having a P-value < 0.05 and confirms the predictor is statistically significant.

4 Section 4: Worked Example Analysis

Below is a sample output table that we will analyze.

Dataset link: [Click Here](#)

Let's apply the theory to the sample output shown in Figure 1.

Model Overview

From the output, we can identify:

- **Observations (n):** 200
- **Independent Variables (k):** 3 (X Variable 1, X Variable 2, X Variable 3)

The regression model being estimated is:

$$y = \beta_0 + \beta_1(\text{X Variable 1}) + \beta_2(\text{X Variable 2}) + \beta_3(\text{X Variable 3}) + \epsilon$$

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.950048046							
R Square	0.90259129							
Adjusted R Square	0.90110034							
Standard Error	1.661695147							
Observations	200							
ANOVA								
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>			
Regression	3	5014.78272	1671.59424	605.380131	8.134E-99			
Residual	196	541.2012295	2.76123076					
Total	199	5555.98395						
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	4.625124079	0.307501165	15.040997	1.6827E-34	4.01868836	5.2315598	4.01868836	5.2315598
X Variable 1	0.05444578	0.001375188	39.5915245	1.8929E-95	0.05173372	0.05715784	0.05173372	0.05715784
X Variable 2	0.107001228	0.008489563	12.6038566	4.6021E-27	0.09025861	0.12374384	0.09025861	0.12374384
X Variable 3	0.000335658	0.005788056	0.05799148	0.9538145	-0.0110792	0.01175052	-0.0110792	0.01175052

Figure 1: Sample regression output from Excel.

Analysis of Section 1: Regression Statistics

- **R Square (0.90259):** This is a strong fit. It indicates that 90.26% of the total variation in the dependent variable (y) is explained by the three independent variables (x_1, x_2, x_3).
- **Standard Error (1.66169):** On average, the model's predictions are off by about 1.66 units of y .

Analysis of Section 2: ANOVA (Overall Model Significance)

This section tests whether the model as a whole is useful.

- **Hypotheses:**

$$H_0 : \beta_1 = \beta_2 = \beta_3 = 0$$

$$H_a : \text{At least one } \beta_j \neq 0$$

- **F-statistic (605.38):** This is the test statistic.
- **Significance F (8.134E-99):** This is the p-value for the F-statistic. The number is 8.134×10^{-99} , which is incredibly small (practically zero).

Conclusion: Since the p-value (Significance F) is much smaller than any standard alpha level (e.g., $\alpha = 0.05$), we **reject the null hypothesis (H_0)**. We conclude that the regression model is **statistically significant**. At least one of our X variables is a meaningful predictor of y .

Analysis of Section 3: Coefficients (Individual Variable Significance)

This section tests each predictor individually. The null hypothesis for each row is $H_0 : \beta_j = 0$.

- **Intercept:**
 - **P-value (1.6827E-34):** This is $\ll 0.05$.
 - **Conclusion:** The intercept is statistically significant.
- **X Variable 1:**

- **P-value (1.8929E-95):** This is $\ll 0.05$.
- **Conclusion: Significant.** X Variable 1 is a statistically significant predictor of y .
- **X Variable 2:**
 - **P-value (4.6021E-27):** This is $\ll 0.05$.
 - **Conclusion: Significant.** X Variable 2 is a statistically significant predictor of y .
- **X Variable 3:**
 - **P-value (0.9538):** This is $\gg 0.05$.
 - **Conclusion: Not Significant.** We fail to reject the null hypothesis. This variable does not add statistically significant predictive power to the model (given that X1 and X2 are already included).
 - **Confidence Interval Check:** Notice the "Lower 95%" (-0.0117) and "Upper 95%" (0.0117) for this variable. This interval *contains zero*, which confirms the t-test result: the coefficient is not statistically different from zero.

Final Summary of Example

Based on this output, our overall model is an excellent fit (high R^2) and is statistically significant (low Significance F). However, when looking at the individual predictors, we find that **X Variable 1** and **X Variable 2** are significant, but **X Variable 3** is not (hence we can omit this while making a prediction).

If we were to write the equation for the model, it would be:

$$\hat{y} = 4.6251 + 0.0544 \times (\text{X Variable 1}) + 0.1070 \times (\text{X Variable 2}) + 0.0003 \times (\text{X Variable 3})$$